

TEAU 1600 – Electrical I (4 Credits)

Course Description

The Electrical I course provides theory and hands-on instruction on automotive electrical systems while following the program standards set forth by Automotive Service Excellence Education Foundation at the master level.

Course Objectives

- Maintain vehicle safety through safe electrical maintenance and repairs.
- Identify and repair electrical/electronic series, parallel, and series-parallel circuits using principles of electricity (Ohm's Law).
- Identify and repair the causes and effects from shorts, grounds, opens, and resistance problems in electrical/electronic circuits.
- Identify and demonstrate proper use of a digital multimeter (DMM) when measuring source voltage, voltage drop (including grounds), current flow and resistance.
- Identify and repair automotive electrical/electronic systems including battery systems, charging systems starting systems, and lighting systems.

Course Outline

- Intro to Automotive Electricity and Electronic Systems
- Electrical and Electronic Theories and Components
- Wiring, Circuit Diagrams, Batteries, and Starting Systems
- Charging, Lighting, and Intro to Body Computers
- Computer Inputs, Sensor Diagnostics, and Communications
- Advanced Lighting and Instrumentation
- Electrical Accessories and Infotainment Systems
- Passive Restraint and Alternative Power Vehicles

Textbook & Reading Materials

Today's Technician: Automotive Electricity and Electronics, Classroom and Shop Manual Pack, 8th ed. (ISBN: 9780357766385)

Assignments & Assessments

Orientation

Orientation Acknowledgement

Electrical I Syllabus 2021-22

Remind Txt Group

Automotive Student OE Instructions

Right to Know and Safety Agreement review

Student Information Sheet

Cleaning Expectations

COVID-19 Policies and Procedures Agreement review

Electrical I Lab Assignment Checklist review

Southwest Technical College Automotive Video Playlist

Student Tool and Equipment Use Waiver

Cell Phone

Instructions

Digital Lab Explanation

Module Breakdown

Module 1 Labs

Module 2 Labs

Module 3 Labs

Module 4 Labs

Module 5 Labs

Module 6 Labs

Module 7 Labs

Module 8 Labs

Cleaning Lab Module

Cleaning Lab 1

Cleaning Lab 2

Cleaning Lab 3

Cleaning Lab 4

Cleaning Lab 5

Read It: Chapter 01 Introduction to Automotive Electrical and Electronic Systems

Chapter 01 Multiple Choice Questions

Chapter 01 Fill-in-the-Blank Questions

Read It: Chapter 01 Safety

Chapter 01 ASE-Style Review Questions

Read It: Chapter 02 Special Tools and Procedures

Using Ohms Law

Series Circuits

Parallel Circuits

Series-Parallel Circuits

Applying Ohms Law

Using the Voltmeter

Voltage Drop Testing

Using an Ohmmeter

Chapter 02 ASE-Style Review Questions

Chapter 02 ASE Challenge Questions

Amperage Testing

Fluke Meter

Lab 1 Research vehicle service info

Lab 2 Use wiring diagrams to trace electrical circuits

Lab 3 Demonstrate proper use of DMM when measuring voltage

Lab 4 Measure source voltage and perform voltage drop test

in a light circuit (Parallel circuit)

Lab 5 Measure current flow in a light circuit and component (Series circuit)

Lab 6 Demonstrate knowledge of the causes and effects from short to grounds

Lab 7 Perform battery state of charge test

Lab 8 Confirm proper battery capacity for vehicle, perform battery capacity and load test

Lab 9 Perform slow/fast battery charge

Read It: Chapter 02 Basic Theories

Chapter 02 Fill-in-the-Blank Questions

Chapter 02 Multiple Choice Questions

Read It: Chapter 03 Electrical and Electronic Components

Chapter 03 Multiple Choice Questions

Chapter 03 Fill-in-the-Blank Questions

Read It: Chapter 03 Basic Electrical Troubleshooting and Service

Chapter 03 ASE-Style Review Questions

Chapter 03 ASE Challenge Questions

Testing for Opens

Testing for Shorts

Testing Relays

Testing Switches

Using the Lab Scope

Using a Scan Tool

Lab 10 Demonstrate knowledge of electrical circuits using (OHMS LAW)

Lab 11 Demonstrate proper use of DMM when measuring current flow

Lab 12 Demonstrate proper use of DMM when measuring resistance

Lab 13 Check electrical circuits with a test light

Lab 14 Check electrical circuit fuse a fused jump wire

Lab 15 Demonstrate knowledge of the causes and effects from short to grounds

Lab 16 Inspect and clean battery, fill cells, check cables, connectors, clamps and hold downs

Lab 17 Inspect and test starter relays and solenoids

Lab 18 Perform starting system output test

Lab 19 Perform charging circuit voltage drop tests

Checkpoint Meeting Module 2

Read It: Chapter 04 Wiring and Circuit Diagrams

Reading Wiring Diagrams

Chapter 04 Multiple Choice Questions

Chapter 04 Fill-in-the-Blank Questions

Read It: Chapter 04 Wiring Repair and Reading Wiring Diagrams

Wire Stripping

Crimping a Wire Connection

Soldering Wires

How to Solder Two Wires Together

Soldering Copper Wire

Chapter 04 ASE-Style Review Questions

Chapter 04 ASE Challenge Questions

Read It: Chapter 05 Automotive Batteries

Chapter 05 Multiple Choice Questions

Chapter 05 Fill-in-the-Blank Questions

Read It: Chapter 05 Battery Diagnosis and Service

Testing Batteries

Battery Testing

How to Test and Replace a Bad Car Battery

Chapter 05 ASE-Style Review Questions

Chapter 05 ASE Challenge Questions

Read It: Chapter 06 Starting Systems and Motor Designs

Starting Systems

Starter Motor Construction

DC Motor: How It Works

Brushless DC Motor: How It Works

Chapter 06 Multiple Choice Questions

Chapter 06 Fill-in-the-Blank Questions

Read It: Chapter 06 Starting System Diagnosis and Service

How to Diagnose and Replace a Starter

Voltage Drop Testing a Starter Motor

Starting System Diagnostics

Chapter 06 ASE-Style Review Questions

Chapter 06 ASE Challenge Questions

Lab 20 Use wiring diagrams during diagnosis of electronic circuit

Lab 21 Use wiring diagrams during diagnosis of electronic circuit

Lab 22 Perform battery state of charge test

Lab 23 Confirm proper battery capacity for vehicle, perform battery capacity and load test

Lab 24 Perform slow/fast battery charge

Lab 25 Perform starter current draw test

Lab 26 Perform starter current draw test

Lab 27 Inspect and test starter relays and solenoids

Lab 28 Remove and install starter in vehicle

Lab 29 Disassemble, inspect, and reassemble starter motor,

perform bench test

Checkpoint Meeting Module 3

Read It: Chapter 07 Charging Systems

Types of Hybrid Vehicle Power Systems

How does an Alternator Work?

How Does Diode Work?

How AC is Turned into DC

Chapter 07 Multiple Choice Questions

Chapter 07 Fill-in-the-Blank Questions

Read It: Chapter 07 Charging System Testing and Service Charging Systems Test

How to Test an Alternator

Using an Ammeter

Charging System Diagnostics

Chapter 07 ASE-Style Review Questions

Chapter 07 ASE Challenge Questions

Read It: Chapter 11 Lighting Circuits

Head Light Circuits

Understanding the Difference between Projector and Reflect

or Headlights

Multibeam Headlamp Technology

Matrix LED Headlight Technology Explained

How Laser Headlights Work

High-Intensity Discharge (HID) Bulbs Explained

How Automatic Headlights Work

How Automatic High Beams Work

Understanding Halogen Headlight Bulbs

Chapter 11 Multiple Choice Questions

Chapter 11 Fill-in-the-Blank Questions

Read It: Chapter 11 Lighting Circuits Diagnostics and Repair

Chapter 11 ASE-Style Review Questions

Head Lights

Lab 30 Measure current flow in a light circuit and component

(Parallel circuit)

Lab 31 Inspect and clean battery, fill cells, check cables, connectors, clamps and hold downs

Lab 32 Perform starter current draw test

Lab 33 Perform charging system output test

Lab 34 Diagnose charging system for the cause of undercharge condition

Lab 35 Diagnose charging system for the cause of no charge

condition

Lab 36 Diagnose charging system for the cause of

overcharge condition

Lab 37 Inspect, adjust, or replace generator (alternator) drive

belts, check pulleys & tensioners for wear and alignment.

Lab 38 Remove, inspect, and install generator (alternator)

Lab 39 Perform charging circuit voltage drop tests

Lab 40 Disassemble, inspect, and reassemble generator (alternator), perform bench tests

Lab 41 Inspect headlamps and sockets, replace as needed

Checkpoint Meeting Module 4

Read It: Chapter 08 Introduction to the Body Computer

Solenoid Basics Explained

How Relays Work

Actuators Explained

Chapter 08 Multiple Choice Questions

Chapter 08 Fill -in-the-Blank Questions

Read It: Chapter 08 Body Computer System Diagnosis

Testing the BCM and Power Ground Circuits

Flashing the BCM

Chapter 08 ASE-Style Review Questions

Chapter 08 ASE Challenge Questions

Read It: Chapter 09 Computer Inputs

How do Hall Effect Sensors Work?

Chapter 09 Multiple Choice Questions

Chapter 09 Fill-in-the-Blank Questions

Read It: Chapter 09 Sensor Diagnostic Routines

Diagnostic Strategies (8-Step Process)

Chapter 09 ASE-Style Review Questions

Chapter 09 ASE Challenge Questions

Read It: Chapter 10 Vehicle Multiplexing Diagnostics

Automotive Computer Networks

CAN Bus: Troubleshooting Common Problems

Chapter 10 Multiple Choice Questions

Chapter 10 Fill -in-the-Blank Questions

Read It: Chapter 10 Vehicle Communication Networks

Diagnosing Network Communication Problems

Chapter 10 ASE-Style Review Questions

Chapter 10 ASE Challenge Questions

Lab 42 Check electrical circuits with a test light

Lab 43 Check continuity and resistance in electrical circuit components

Lab 44 Check continuity and resistance in electrical circuit components

Lab 45 Check electrical circuit fuse a fused jump wire

Lab 46 Replace electrical connectors and terminals

Lab 47 Maintain or restore electronic memory functions

Lab 48 Inspect and test switches, connectors, and wires of started control circuits

Checkpoint Meeting Module 5

Read It: Chapter 12 Instrumentation and Warning Lamp Gauges

Chapter 12 Multiple Choice Questions

Chapter 12 Fill -in-the-Blank Questions

Read It: Chapter 12 Instrumentation and Warning Lamp System Diagnosis and Repair

Chapter 12 ASE-Style Review Questions

Chapter 12 ASE Challenge Questions

Lab 49 Measure source voltage and perform voltage drop test in a light circuit (Series circuit)

Lab 50 Measure source voltage and perform voltage drop test in a light circuit (Parallel circuit)

Lab 51 Inspect and test fusible links, circuit breakers, and fuses

Lab 52 Inspect, test, repair and/or replace components in electrical systems, determine needed action

Lab 53 Diagnose the causes of brighter-than-normal, intermittent, dim, or no light operation

Lab 54 Diagnose the causes of brighter-than-normal, intermittent, dim, or no light operation

Lab 55 Inspect turn signal/hazard flasher lamps and sockets, replace as needed

Lab 56 Inspect all exterior lamps and sockets (except headlamps and turn signals), replace as needed

Lab 57 Aim headlights

Lab 58 Demonstrate knowledge of the causes and effects from shorts to grounds

Checkpoint Meeting Module 6

Read It: Chapter 13 Accessories

Blower Motor Control

Multispeed Blower Circuit Operation

Chapter 13 Multiple Choice Questions

Chapter 13 Fill-in-the-Blank Questions

Read It: Chapter 13 Electrical Accessories Diagnosis and Repair

Chapter 13 ASE-Style Review Questions

Chapter 13 ASE Challenge Questions

Read It: Chapter 14 Radio Frequency, Infotainment, and Connected Vehicle Technology

Vehicle Security Systems

Chapter 14 Multiple Choice Questions

Chapter 14 Fill-in-the-Blank Questions

Read It: Chapter 14 Servicing Radio Frequency and Infotainment Systems

Check Vehicle Security Systems

Chapter 14 ASE-Style Review Questions

Chapter 14 ASE Challenge Questions

Lab 59 Measure key-off battery drain (parasitic draw)

Lab 60 Measure key-off battery drain (parasitic draw)
 Lab 61 Diagnose the cause of excessive key-off battery on (parasitic draw) determine
 Lab 62 Inspect and test fusible links, circuit breakers, and fuses
 Lab 63 Repair wiring harness, perform solder repair of electrical wiring
 Lab 64 Identify electrical system components
 Lab 65 Identify electronic modules, security systems, radios, and other accessories
 Lab 66 Inspect, adjust, or replace generator (alternator) drive belts, check pulleys & tensioners for wear and alignment.
 Lab 67 Inspect interior lamps and sockets, replace as needed
 Lab 68 Demonstrate knowledge of the causes and effects from shorts to grounds
 Checkpoint Meeting Module 7
 Read It: Chapter 15 Passive Restraint and Occupant Safety Systems
 Chapter 15 Multiple Choice Questions
 Chapter 15 Fill-in-the-Blank Questions
 Read It: Chapter 15 Servicing Passive Restraint and Occupant Safety Systems
 Disarming an Airbag System
 Chapter 15 ASE-Style Review Questions
 Chapter 15 ASE Challenge Questions
 Read It: Chapter 16 Advanced Driver Assistance Systems
 Chapter 16 Multiple Choice Questions
 Chapter 16 Fill-in-the-Blank Questions
 Read It: Chapter 16 Servicing ADAS
 Chapter 16 ASE-Style Review Questions
 Chapter 16 ASE Challenge Questions
 Read It: Chapter 17 HEV, EV, and Alternative Power Sources

Electric Vehicle Designs
 Hybrid Vehicles
 Types of Hybrid Vehicle Power Systems
 General Hybrid Electric Vehicle Safety
 Chapter 17 Multiple Choice Questions
 Chapter 17 Fill-in-the-Blank Questions
 Read It: Chapter 17 Hybrid and High-Voltage System Service
 Preparing the HEV for Service
 Chapter 17 ASE-Style Review Questions
 Chapter 17 ASE Challenge Questions
 Lab 69 Inspect, test, repair and/or replace components in electrical systems, determine needed action
 Lab 70 Jump-start a vehicle using jumper cables and a booster battery or auxiliary power supply
 Lab 71 Identify high voltage circuit of electric or hybrid electric vehicles and related safety precautions
 Lab 72 Identify hybrid vehicles auxiliary (12v) battery service, repair and test procedures
 Lab 73 Differentiate between electrical and mechanical problems that cause slow or no-crank
 Lab 74 Demonstrate knowledge of an automatic idlesstop/start-stop system
 Lab 75 Aim headlights
 Lab 76 Identify system voltage and safety precautions for HID lighting systems
 Lab 77 Demonstrate knowledge of the causes and effects from shorts to grounds
 Checkpoint Meeting Module 8
 End of Course Survey
 Electrical Systems Competency Profile (2021)
 Electrical 1 Review
 Electrical 1 Final Exam

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Mo, Tu, W, Th

8:00 AM - 12:00 AM

1:00 PM - 5:00 PM

Friday

8:00 AM - 12:00 AM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

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Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

Course Grading: Students must achieve 80% (B-) or higher to pass graded work. Incomplete assignments must be redone to meet the required standards. Guidelines, rules, and expectations for completing assignments are provided in each course.

High School Power School Grades: Quarter student grades will be determined by student progress percentage. Faculty will use the higher percentage of either 1) quarter progress, or 2) cumulative progress for the current training plan year.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

Class attendance is required during your scheduled time. Cell phones for many have become a distraction. When you are in class or lab we encourage you to keep your cell phones put away in a secure location. If you use ear buds we ask that you only use one so you can still hear the things going on around you. If you are using your phone for things other than school related items, instructors will ask you to put them away. Please follow the direction of your instructors. Those who have been asked to refrain from using your cell phone and fail to do so will be asked to meet with the Director of Transportation and student services will be notified. The program is designed to provide the student with as much hands-on work as possible. In the automotive industry you may be required to lift heavy objects and stand for hours at a time to complete work required. Technicians deal with chemicals and materials which require caution, these will be identified in the Right to Know Agreement provided to you. You will also be required to use computers to track and complete work.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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